

Algorithm: BERT-based Transformer for classifying Fake NewsDataset: [LIAR Dataset]: [Dataset](#)Reference Notebook: [Notebook](#)Reference Dataset Paper: [Paper](#)

Task	Actions Taken	Rationale	Difficulty Level (1-10)	Time Taken
Data Understanding	Explored the LIAR dataset, examined features, and identified the target variable (Labels).	For getting insight into the structure of the dataset and verifying the suitability for classification.	4	40 min
Data Preprocessing	Tokenized text data using BERT tokenizer and encoded labels. Divided dataset into training, validation, and test splits.	Tokenization makes it easy for BERT to process textual inputs efficiently. Ensure proper model training, evaluation, and testing for generalizability.	5	50 min
Implemented Transformer using BERT (Baseline Model Implementation)	Used a pre-trained BERT model (bert-base-uncased) and fine-tuned on the LIAR dataset.	BERT is highly efficient in capturing contextual meaning in text classification problems. Adjusted learning rate (2e-5), batch size (16), and epochs (5) for optimal performance.	8	2.5 hours
Hyperparameter Tuning	Finding the appropriate parameters ensures quick convergence and enhanced accuracy.	To assess performance and identify where improvement needs to be made.	6	1 hour
Model Evaluation	Computed accuracy, precision, recall, and F1-score. Verified	To measure performance and confirm the correctness of the	4	30 min

	misclassified samples, particularly uncertain statements.	implementation. Helps to identify biases and potential improvements.		
Conclusion & Analysis	Summarized results, suggested using RoBERTa or data augmentation for improvement.	Guides future enhancements and model updates.	3	30 min

1. Business Understanding

Objective: This work focuses on classifying false news using the LIAR dataset with a BERT-based transformer model. The model is trained to distinguish different levels of truthfulness in statements. Some of the challenging parts are handling imbalanced classes, selecting the optimal hyperparameters, and to interpret the model's predictions.

2. Data Understanding:

The LIAR dataset consists of truth-labeled political statements in different categories of truthfulness. This dataset contains textual statements, metadata, and the respective truth labels. Data Preprocessing consists of tasks such as text tokenization and label encoding to make the data BERT-compatible.

The LIAR dataset is a benchmark dataset for fake news detection. It includes 12,836 labeled short political statements collected from Politifact, a fact-checking web site. All statements are annotated into one of six levels of truthfulness:

- "pants-fire" (completely false)
- "false"
- "barely-true"
- "half-true"
- "mostly-true"
- "true"

3. Data Preparation:

- Utilized BertTokenizer from transformers library for applied tokenization.
- Translated labels into PyTorch compatibility.
- Split dataset into training, validation, and test sets.

4. Modeling - Baseline Implementation (custom):

Implementation Details:

- Utilized bert-base-uncased from Hugging Face.
- Added a multi-class classification head.
- Used AdamW optimizer with weight decay.
- Learning Rate: $2e-5$
- Batch Size: 16
- Epochs: 3
- Loss Function: Cross-Entropy Loss

5. Evaluation:

Number of epochs: 3

Starting training...

 [1926/1926 46:21, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	1.693500	1.710560	0.239875	0.148599	0.239875	0.177007
2	1.654800	1.676400	0.267913	0.270981	0.267913	0.253034
3	1.281800	1.851282	0.267913	0.274511	0.267913	0.265934

Evaluating the model...

 [81/81 00:35]

```
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'eval_f1': 0.2659339830544386,  
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'eval_steps_per_second': 2.226,  
'epoch': 3.0}
```

Number of epochs: 8

Starting training...

[5136/5136 2:04:31, Epoch 8/8]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	1.260800	2.063118	0.235981	0.251018	0.235981	0.227488
2	0.859000	2.451596	0.251558	0.257262	0.251558	0.239026
3	0.746400	2.900211	0.245327	0.253179	0.245327	0.243784
4	0.409600	3.577487	0.255452	0.265937	0.255452	0.249779
5	0.126800	4.418734	0.255452	0.262929	0.255452	0.248868
6	0.109500	5.177073	0.254673	0.261092	0.254673	0.251116
7	0.132800	5.783558	0.260125	0.264197	0.260125	0.256022
8	0.060500	5.953994	0.252336	0.255888	0.252336	0.250116

Evaluating the model...

[81/81 00:35]

```
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'eval_samples_per_second': 35.285,  
'eval_steps_per_second': 2.226,  
'epoch': 8.0}
```

Observations:

- The model performed well on fact-based statements but struggled with ambiguous statements. There is potential for improvement by adding external knowledge sources or using larger transformer models.

6. Summary

- Fine-tuned BERT model on LIAR dataset.
- Carried out preprocessing through tokenization and encoding of the labels.
- Evaluating the model's performance with a set of evaluation metrics.
- Observed where the model had to get better, such as support for uncertain sentences and data augmentation.

Next Steps:

- Experiment with larger models such as RoBERTa.
- Applying data augmentation methods to generalise better.